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Grade 10.00 out of 10.00 (100%)

Question 1

Correct

Mark 2.50 out of 2.50

In a machine learning monitoring system, let X denote the number of high-priority anomalies detected in a batch. Suppose X follows the probability mass function

$$p(x) = cx^2, x = 1, 2, 3$$

Which of the following correctly gives the values of c , $E[X]$, $\text{Var}(X)$, and $E[X(X-1)^2]$?

- ☒ A. $c = \frac{1}{14}, E[X] = \frac{18}{7}, \text{Var}(X) = \frac{19}{49}, E[X(X-1)^2] = \frac{58}{7}$ ✓
- ☐ B. $c = \frac{1}{12}, E[X] = \frac{9}{4}, \text{Var}(X) = \frac{15}{16}, E[X(X-1)^2] = \frac{29}{4}$
- ☐ C. $c = \frac{1}{14}, E[X] = \frac{16}{7}, \text{Var}(X) = \frac{18}{49}, E[X(X-1)^2] = \frac{48}{7}$
- ☐ D. $c = \frac{1}{10}, E[X] = \frac{11}{5}, \text{Var}(X) = \frac{14}{25}, E[X(X-1)^2] = \frac{24}{5}$

Your answer is correct.

The correct answer is: $c = \frac{1}{14}, E[X] = \frac{18}{7}, \text{Var}(X) = \frac{19}{49}, E[X(X-1)^2] = \frac{58}{7}$

Question 2

Correct

Mark 2.50 out of 2.50

In an AI-based quality control system, a manufacturer's memory chips are inspected automatically. Each chip has a 1% probability of being defective. A batch contains 20 chips.

If X is the number of defective chips in the batch, which of the following gives the correct value of:

- the probability that at most one chip is defective, and
- the mean and variance of the number of non-defective chips? (hint: use binomial Dist.)

- ☒ A. $P(X \leq 1) = 0.98314, \mu = 19.8, \text{Var}(X) = 0.198$ ✓
- ☐ B. $P(X \leq 1) = 0.81791, \mu = 0.2, \text{Var}(X) = 0.198$
- ☐ C. $P(X \leq 1) = 0.98314, \mu = 20, \text{Var}(X) = 0.01$
- ☐ D. $P(X \leq 1) = 0.99000, \mu = 19.8, \text{Var}(X) = 0.99$

Your answer is correct.

The correct answer is: $P(X \leq 1) = 0.98314, \mu = 19.8, \text{Var}(X) = 0.198$

Question 3

Correct

Mark 2.50 out of 2.50

In an AI-based predictive maintenance application, the lifetime T (in months) of a light bulb is modeled by the probability density function

$$f(t) = \begin{cases} k - \frac{t}{450}, & 0 \leq t \leq 30 \\ 0, & \text{otherwise} \end{cases}$$

Which of the following gives the correct values of the constant k and the probability that the bulb lasts at least 15 months?

- ☐ A. $k = \frac{1}{30}, P(T \geq 15) = \frac{1}{2}$
- ☐ B. $k = \frac{1}{15}, P(T \geq 15) = \frac{1}{2}$
- ☐ C. $k = \frac{2}{15}, P(T \geq 15) = \frac{1}{3}$
- ☒ D. $k = \frac{1}{15}, P(T \geq 15) = \frac{1}{4}$ ✓

Your answer is correct.

The correct answer is: $k = \frac{1}{15}, P(T \geq 15) = \frac{1}{4}$

Question 4

Correct

Mark 2.50 out of 2.50

In a data science model for anomaly-score analysis, the random variable X follows a normal distribution $N(\mu, \sigma)$. Suppose

$$P(X < 66) = 0.0359, P(X > 81) = 0.1151$$

Using the given standard normal table, determine μ , σ , and $P(69 < X < 83)$.

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APPENDIX A

NEGATIVE z Scores

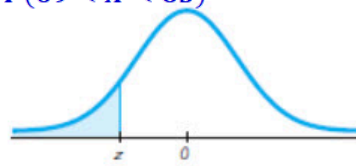
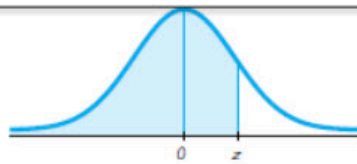


TABLE A-2 Standard Normal (z) Distribution: Cumulative Area from the LEFT

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.50 and lower	.0001									
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641



POSITIVE z Scores

TABLE A-2 (continued) Cumulative Area from the LEFT

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.50 and up	.9999									

- ☐ A. $\mu = 70, \sigma = 6, P(69 < X < 83) = 0.7251$
☐ B. $\mu = 75, \sigma = 3, P(69 < X < 83) = 0.8849$
☐ C. $\mu = 72, \sigma = 5, P(69 < X < 83) = 0.9452$
☒ D. $\mu = 75, \sigma = 5, P(69 < X < 83) = 0.7673$ ✓

Your answer is correct.

The correct answer is: $\mu = 75, \sigma = 5, P(69 < X < 83) = 0.7673$